

Project: Coffee Shop Sales Visualization

Problem Statement

A coffee shop owner wants to better understand the sales performance of different coffee drinks using data visualization. The owner has collected information about the number of cups sold and the price of each coffee. As a Data Analyst, your task is to use NumPy, Pandas, Matplotlib, and Seaborn to organize the data, create visualizations, and generate meaningful business insights.

Objective

Create a Python program using NumPy, Pandas, Matplotlib, and Seaborn to analyze coffee shop sales data and present the results through different charts and graphs.

Requirements

- Import NumPy, Pandas, Matplotlib, and Seaborn.
- Create NumPy arrays for Coffee Name, Cups Sold, and Price per Cup.
- Create a Pandas DataFrame.
- Create a Revenue column (Cups Sold × Price per Cup).
- Display the DataFrame.
- Create Bar, Pie, Line, Histogram, Scatter, Seaborn Bar Plot, Box Plot, and Correlation Heatmap.
- Display Most Sold Coffee, Most Expensive Coffee, Average Coffee Price, and Total Revenue.

Sample Data

Coffee	Cups Sold	Price (■)
Espresso	120	120
Latte	95	180
Cappuccino	150	160
Mocha	80	200
Cold Coffee	110	150

Concepts Covered

- Importing Libraries
- NumPy Arrays
- Pandas DataFrame
- Creating New Columns
- Matplotlib
- Seaborn
- Bar Chart
- Pie Chart
- Line Chart
- Histogram

- Scatter Plot
- Box Plot
- Heatmap
- Business Data Analysis

Expected Learning Outcomes

- Create NumPy arrays and Pandas DataFrames.
- Perform business data analysis.
- Generate charts using Matplotlib and Seaborn.
- Interpret business insights through visualizations.